

UCSH-604: Deep Learning (Elective - I)

McCulloch-Pitts Neuron:

Input: $x_1, x_2, \dots, x_n \in \{0, 1\}$

Output : $y \in \{0, 1\}$

Model : $\text{if } \sum_{i=1}^n x_i \geq \theta \Rightarrow 1$
 $\Rightarrow 0, \text{ otherwise}$

Perception:

Input: $x_1, x_2, \dots, x_n \in \mathbb{R}$

Our prob: $y \in \{0, 1\}$

Parameters: $w_0, w_1, w_2, \dots, w_n$ ($n+1$)

Model :

$$\sum_{i=0}^n w_i x_i \geq 0 \quad \left. \begin{array}{l} y = 1 \\ \rightarrow y = 0, \\ \text{otherwise} \end{array} \right\} \quad (x_0 = 1)$$

Sai

$$L(w) = L(w_0) + \frac{L'(w_0)}{1!} (w - w_0)$$

$$y = mx + c \quad \begin{array}{l} (x_0, y_0) \\ \underline{\underline{(x_1, y_1)}} \end{array}$$

$$y_0 = mx_0 + c$$

$$y_1 = mx_1 + c$$

$$y_1 - y_0 = m(x_1 - x_0)$$

$$y_1 = y_0 + m(x_1 - x_0)$$

Sai

Gradient Descent

Case (i): One Sigmoid Neuron, One data point, Single feature.

$$\hat{y}_1 = \sigma(x_1; w, b)$$

$$= \frac{1}{1 + e^{-(wx_1 + b)}}$$



$$L(w, b) = \frac{1}{2} (\hat{y}_1 - y_1)^2$$

We need to find $\frac{\partial L}{\partial w}$ and $\frac{\partial L}{\partial b}$.

$$\text{Let } z = wx_1 + b$$

$$k = \frac{1}{1 + e^{-z}}$$

$$\Rightarrow L = \frac{1}{2} (\hat{y}_1 - y_1)^2 = \frac{1}{2} (k - y_1)^2$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial k} \times \frac{\partial k}{\partial z} \times \frac{\partial z}{\partial w}$$

$$\frac{\partial L}{\partial k} = \frac{1}{2} \times 2 \times (k - y_1) \times 1 = k - y_1 = \hat{y}_1 - y_1$$

$$\frac{\partial k}{\partial z} = \frac{-1}{(1 + e^{-z})^2} (-e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2} = \frac{1}{(1 + e^{-z})} \times \frac{e^{-z}}{(1 + e^{-z})}$$

$$\text{Consider } 1 - k = 1 - \frac{1}{1 + e^{-z}} = \frac{e^{-z}}{1 + e^{-z}}$$

$$\Rightarrow \frac{\partial k}{\partial z} = k(1 - k) = \hat{y}_1(1 - \hat{y}_1)$$

$$\frac{\partial z}{\partial w} = \frac{\partial}{\partial w} (wx_1 + b) = x_1$$

$$\Rightarrow \frac{\partial L}{\partial w} = (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) x_1$$

Similarly $\frac{\partial L}{\partial b} = \frac{\partial L}{\partial k} \times \frac{\partial k}{\partial Z} \times \frac{\partial Z}{\partial b}$

$$\frac{\partial Z}{\partial b} = \frac{\partial}{\partial b} (wx_1 + b) = 1$$

$$\Rightarrow \frac{\partial L}{\partial b} = (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) \times 1$$

$$= (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1)$$

Case (ii): One Sigmoid Neuron, n features,
One data point (x_1, y_1) , where
 $x_1 = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, $y_1 \in [0, 1]$

$\Rightarrow$ Parameters: $w_1, w_2, \dots, w_n, b$

$$\Rightarrow \hat{y}_1 = \frac{1}{1 + e^{-(w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b)}}$$

Let $W = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$, then $\hat{y}_1 = \frac{1}{1 + e^{-(W \cdot x_1 + b)}}$

$$L(W, b) = \frac{1}{2} (\hat{y}_1 - y_1)^2$$

We need to find $\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \dots, \frac{\partial L}{\partial w_n}, \frac{\partial L}{\partial b}$

Let $Z = W \cdot x_1 + b$

$$k = \frac{1}{1 + e^{-z}}$$

$$L(w, b) = \frac{1}{2} (k - y_1)^2$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial k} \times \frac{\partial k}{\partial z} \times \frac{\partial z}{\partial w_1}$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial k} \times \frac{\partial k}{\partial z} \times \frac{\partial z}{\partial w_2}$$

$\vdots$

$$\frac{\partial L}{\partial w_n} = \frac{\partial L}{\partial k} \times \frac{\partial k}{\partial z} \times \frac{\partial z}{\partial w_n}$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial k} \times \frac{\partial k}{\partial z} \times \frac{\partial z}{\partial b}$$

$$\frac{\partial L}{\partial k} = \frac{1}{2} \times 2 \times (k - y_1) = k - y_1 = \hat{y}_1 - y_1$$

$$\begin{aligned} \frac{\partial k}{\partial z} &= \frac{-1}{(1 + e^{-z})^2} \cdot (e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2} = \frac{1}{1 + e^{-z}} \times \frac{e^{-z}}{1 + e^{-z}} \\ &= \hat{y}_1 (1 - \hat{y}_1) \end{aligned}$$

$$\frac{\partial z}{\partial w_1} = \frac{\partial}{\partial w_1} (w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b) = x_1$$

$$\frac{\partial L}{\partial w_1} = (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) x_1$$

$$\frac{\partial L}{\partial w_2} = (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) x_2$$

$\vdots$

$$\frac{\partial L}{\partial w_n} = (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) x_{n1}$$

$$\frac{\partial L}{\partial b} = (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1)$$

Case (iii): Single Neuron, n inputs / features,
 m data points $((x_1, y_1), (x_2, y_2), \dots, (x_m, y_m))$

$$\Rightarrow X_1 = \begin{bmatrix} x_{11} \\ x_{12} \\ \vdots \\ x_{1n} \end{bmatrix}, X_2 = \begin{bmatrix} x_{21} \\ x_{22} \\ \vdots \\ x_{2n} \end{bmatrix}, \dots, X_m = \begin{bmatrix} x_{m1} \\ x_{m2} \\ \vdots \\ x_{mn} \end{bmatrix}$$

$$\Rightarrow \text{Parameters: } w_1, w_2, \dots, w_n, b, \text{ Let } W = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

$$\therefore \hat{y}_1 = \frac{1}{1 + e^{-(W \cdot x_1 + b)}}, \quad \hat{y}_2 = \frac{1}{1 + e^{-(W \cdot x_2 + b)}}$$

$$\dots \hat{y}_m = \frac{1}{1 + e^{-(W \cdot x_m + b)}}$$

$$L(W, b) = \frac{1}{2} \sum_{i=1}^m (\hat{y}_i - y_i)^2$$

We need to find $\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \dots, \frac{\partial L}{\partial w_n}, \frac{\partial L}{\partial b}$.

$$\text{Let } z_1 = W \cdot x_1 + b, \quad z_2 = W \cdot x_2 + b,$$

$$\dots z_m = W \cdot x_m + b$$

$$k_1 = \frac{1}{1 + e^{-z_1}}, \quad k_2 = \frac{1}{1 + e^{-z_2}}, \quad \dots, \quad k_m = \frac{1}{1 + e^{-z_m}}$$

$$\therefore L(w, b) = \frac{1}{2} \sum_{i=1}^m (k_i - y_i)^2$$

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= \frac{\partial L}{\partial k_1} \times \frac{\partial k_1}{\partial z_1} \times \frac{\partial z_1}{\partial w_1} + \frac{\partial L}{\partial k_2} \times \frac{\partial k_2}{\partial z_2} \times \frac{\partial z_2}{\partial w_1} + \\ &\dots + \frac{\partial L}{\partial k_m} \times \frac{\partial k_m}{\partial z_m} \times \frac{\partial z_m}{\partial w_1} \end{aligned}$$

Let $y = x^2$, $z = \log x$, $u = \underline{\underline{yz}}$

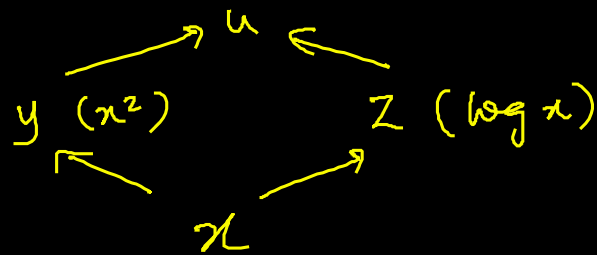
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      graph BT
      x((x)) --> y["y (x^2)"]
      x --> z["z (log x)"]
      y --> u((u))
      z --> u
  
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$$\begin{aligned} \frac{du}{dx} &= \frac{du}{dy} \times \frac{dy}{dx} + \frac{du}{dz} \times \frac{dz}{dx} \\ &= z \times (2x) + y \times \left(\frac{1}{x}\right) \\ &= 2x \log x + x \end{aligned}$$

$$\begin{aligned} f(x) &= x^2 \log x & \frac{df}{dx} &= \frac{x^2}{x} + 2x \log x \\ & & &= x + 2x \log x \end{aligned}$$

Let $y = x^2$, $z = \log x$, $u = \underline{\underline{y + z}}$



$$\frac{du}{dx} = \frac{du}{dy} \times \frac{dy}{dx} + \frac{du}{dz} \times \frac{dz}{dx}$$

$$= 1 \times 2x + 1 \times \frac{1}{x}$$

$$= 2x + \frac{1}{x}$$

$$f(x) = x^2 + \log x \Rightarrow \frac{df}{dx} = 2x + \frac{1}{x}$$

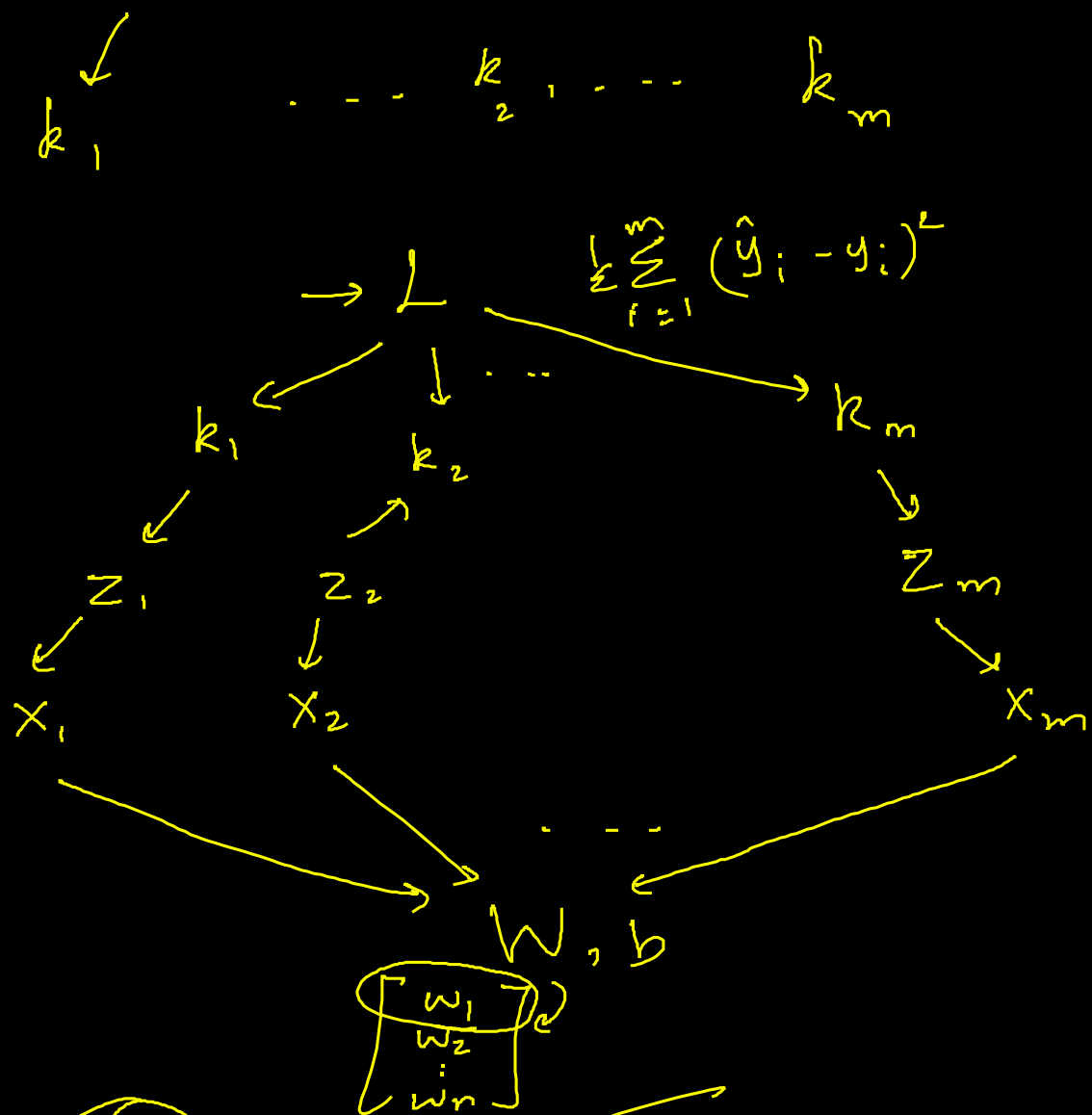
$$L = \frac{1}{2} \sum_{i=1}^m (k_i - y_i)^2$$

$$= \frac{1}{2} \left[(k_1 - y_1)^2 + (k_2 - y_2)^2 + \dots + (k_m - y_m)^2 \right]$$

$$= \frac{1}{2} \left[\left(\frac{1}{1 + e^{-z_1}} - y_1 \right)^2 + \left(\frac{1}{1 + e^{-z_2}} - y_2 \right)^2 + \dots + \left(\frac{1}{1 + e^{-z_m}} - y_m \right)^2 \right]$$

$$z_1 = W \cdot x_1 + b, \quad z_2 = W \cdot x_2 + b, \quad \dots, \quad z_m = W \cdot x_m + b$$

$$= \frac{1}{2} \left[\left(\frac{1}{1 + e^{-(W \cdot x_1 + b)}} - y_1 \right)^2 + \dots + \left(\frac{1}{1 + e^{-(W \cdot x_m + b)}} - y_m \right)^2 \right]$$



$$\frac{\partial L}{\partial w_1} = \left(\frac{\partial L}{\partial k_1} \times \frac{\partial k_1}{\partial z_1} \times \frac{\partial z_1}{\partial w_1} \right) + \frac{\partial L}{\partial k_2} \times \frac{\partial k_2}{\partial z_2} \times \frac{\partial z_2}{\partial w_1} + \dots + \frac{\partial L}{\partial k_m} \times \frac{\partial k_m}{\partial z_m} \times \frac{\partial z_m}{\partial w_1}$$

$$L = \frac{1}{2} \sum_{i=1}^m (\hat{y}_i - y_i)^2 = \frac{1}{2} \times [(k_1 - y_1)^2 + \dots + (k_m - y_m)^2]$$

$$\frac{\partial L}{\partial k_1} = \frac{1}{2} [2 \times (k_1 - y_1) \times 1 + 0 + \dots + 0] = (k_1 - y_1)$$

$$k_1 = \frac{1}{1 + e^{-z_1}} \Rightarrow \frac{\partial k_1}{\partial z_1} = \frac{-1}{(1 + e^{-z_1})^2} \cdot (-e^{-z_1})$$

$$= \frac{1}{(1 + e^{-z_1})} \times \frac{e^{-z_1}}{(1 + e^{-z_1})} = k_1 (1 - k_1)$$

$$= \hat{y}_1 (1 - \hat{y}_1)$$

$$z_1 = W \cdot X_1 + b = w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n} + b$$

$$\frac{\partial z_1}{\partial w_1} = x_{11}$$

$$z_2 = W \cdot X_2 + b = w_1 x_{21} + w_2 x_{22} + \dots + w_n x_{2n} + b$$

$$\frac{\partial z_2}{\partial w_1} = x_{21} \quad ;$$

$\vdots$

$$z_m = W \cdot X_m + b$$

$$= w_1 x_{m1} + w_2 x_{m2} + \dots + w_n x_{mn} + b$$

$$\frac{\partial z_m}{\partial w_1} = x_{m1}$$

$$\frac{\partial L}{\partial w_1} = (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) x_{11} +$$

$$(\hat{y}_2 - y_2) \hat{y}_2 (1 - \hat{y}_2) x_{21} +$$

$\vdots$

$$(\hat{y}_m - y_m) \hat{y}_m (1 - \hat{y}_m) x_{m1}$$

Similarly,

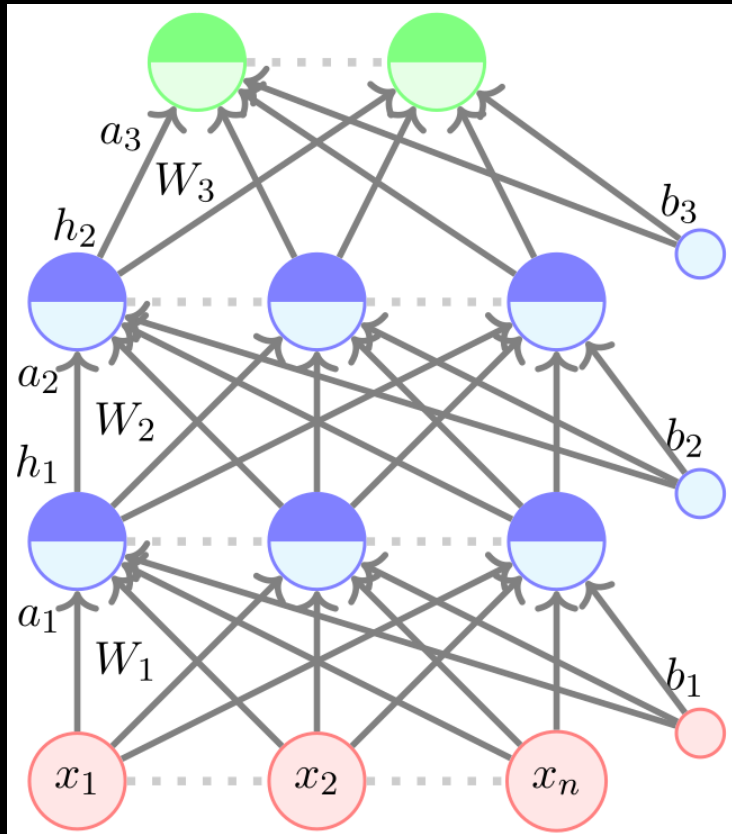
$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \omega_2} = & (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) x_{12} + \\ & (\hat{y}_2 - y_2) \hat{y}_2 (1 - \hat{y}_2) x_{22} + \\ & \vdots \\ & (\hat{y}_m - y_m) \hat{y}_m (1 - \hat{y}_m) x_{m2} \\ & \vdots \end{aligned}$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \omega_n} = & (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) x_{1n} + \\ & (\hat{y}_2 - y_2) \hat{y}_2 (1 - \hat{y}_2) x_{2n} + \\ & \vdots \\ & (\hat{y}_m - y_m) \hat{y}_m (1 - \hat{y}_m) x_{mn} \end{aligned}$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial b} = & (\hat{y}_1 - y_1) \hat{y}_1 (1 - \hat{y}_1) + \\ & (\hat{y}_2 - y_2) \hat{y}_2 (1 - \hat{y}_2) + \\ & \vdots \\ & (\hat{y}_m - y_m) \hat{y}_m (1 - \hat{y}_m) \end{aligned}$$

Sai

Let us now proceed towards multiple layers of Sigmoid Neurons.



→ 3 Layered Network
(Input layer not counted)
(0)

→ a_i - Pre-activation
 h_i - Activation Output

$$\uparrow h_i$$

$$\bigcirc \quad h_i = g(a_i)$$

$$\uparrow a_i$$

→ W_i : weights of layer i

$$W_{ijk} : \begin{matrix} i-1 \text{ to } i \\ j^{\text{th}} \text{ neuron of layer } i \\ k^{\text{th}} \text{ input from layer } i-1 \end{matrix} \quad W_i = \begin{bmatrix} W_{i11} & W_{i12} & \dots & W_{i1n_0} \\ W_{i21} & W_{i22} & \dots & W_{i2n_0} \\ \vdots & \vdots & \ddots & \vdots \\ W_{in_11} & W_{in_12} & \dots & W_{in_1n_0} \end{bmatrix} \rightarrow (n_i \times n_0)$$

n_i : No. of Neurons in layer i .

a_i : Pre-activation Vector to layer i .

b_i : Bias Parameters in layer i .

$$\therefore a_i = \begin{bmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in_i} \end{bmatrix} \quad b_i = \begin{bmatrix} b_{i1} \\ b_{i2} \\ \vdots \\ b_{in_i} \end{bmatrix}$$

Let $x_1 = \begin{bmatrix} x_{11} \\ x_{12} \\ \vdots \\ x_{1n_0} \end{bmatrix}$ be a data point.

Then $a_1 = W_{111}x_{11} + W_{112}x_{12} + \dots + W_{11n_0}x_{1n_0} + b_{11}$

$a_2 = W_{121}x_{11} + W_{122}x_{12} + \dots + W_{12n_0}x_{1n_0} + b_{12}$

$\vdots$

$a_{n_1} = W_{1n_11}x_{11} + W_{1n_12}x_{12} + \dots + W_{1n_1n_0}x_{1n_0} + b_{1n_1}$

$\therefore$ In Matrix form,

$$\begin{bmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{1n_1} \end{bmatrix}_{n_1 \times 1} = \begin{bmatrix} W_{111} & W_{112} & \dots & W_{11n_0} \\ W_{121} & W_{122} & \dots & W_{12n_0} \\ \vdots & \vdots & \ddots & \vdots \\ W_{1n_11} & W_{1n_12} & \dots & W_{1n_1n_0} \end{bmatrix}_{n_1 \times n_0} \begin{bmatrix} x_{11} \\ x_{12} \\ \vdots \\ x_{1n_0} \end{bmatrix}_{n_0 \times 1} + \begin{bmatrix} b_{11} \\ b_{12} \\ \vdots \\ b_{1n_1} \end{bmatrix}_{n_1 \times 1}$$

$a_1 = W_1 x_1 + b_1$

$h_1 = g(a_1) \Rightarrow \begin{bmatrix} h_{11} \\ h_{12} \\ \vdots \\ h_{1n_1} \end{bmatrix}_{n_1 \times 1} = \begin{bmatrix} g(a_{11}) \\ g(a_{12}) \\ \vdots \\ g(a_{1n_1}) \end{bmatrix}_{n_1 \times 1}$

$a_2 = W_2 h_1 + b_2$

$h_2 = g(a_2)$

$$\vdots$$

$$a_{L-1} = W_{L-1} h_{L-2} + b_{L-1} \quad \frac{g}{\text{Logistic}}$$

$$h_{L-1} = g(a_{L-1}) \quad \text{Tanh}$$

$$a_L = W_L h_{L-1} + b_L \quad \text{Linear}$$

$$\hat{y} = h_L = \phi(a_L) \quad \frac{\phi}{\text{softmax}}$$

$\Rightarrow$ for $L=3$ and for (x_i, y_i) linear

$$\hat{y}_i = \phi(W_3 g(W_2 g(W_1 x_i + b_1) + b_2) + b_3)$$

$$L(W, b) = \text{Loss function}(\hat{y}, y) \quad i = 1 \text{ to } m$$

$\rightarrow$ Squared Error

Q : Total No. of Parameters $\rightarrow$ Cross Entropy

BACK PROPAGATION

Consider the following network :

No. of layers = 3

$g = \text{Logistic}$

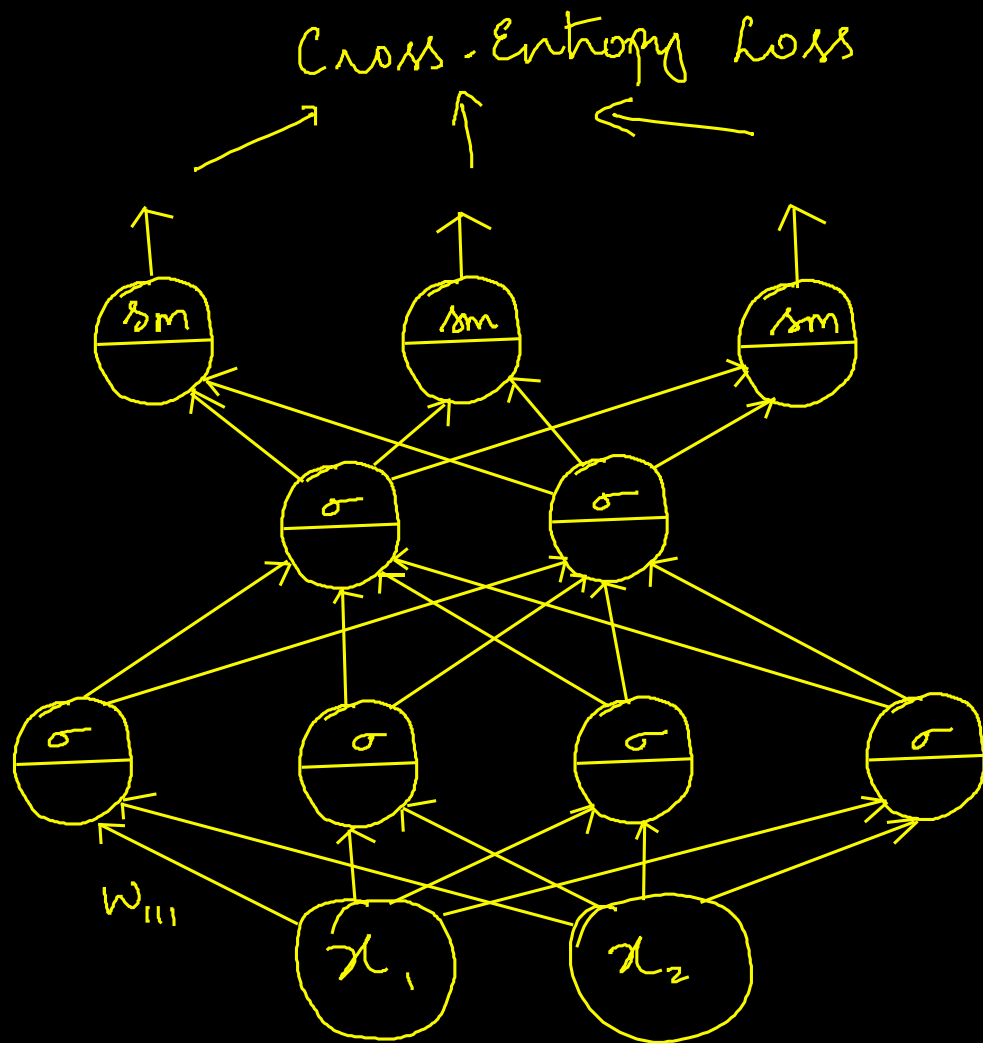
Input layer $n_0 = 2$

First hidden layer = 4 (n_1)

$\phi = \text{softmax}$

Second hidden layer = 2 (n_2) $L = \text{Cross Entropy}$

Output layer = 3 (n_3)



$$W_1: 4 \times 2 \quad W_2: 2 \times 4 \quad W_3: 3 \times 2$$

$$b_1: 4 \times 1 \quad b_2: 2 \times 1 \quad b_3: 3 \times 1$$

$$\text{Let } x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$a_1 = Wx + b_1 \rightarrow h_1 = \sigma(a_1) \rightarrow$$

$$a_3 = W_3 h_2 + b_3 \leftarrow h_2 = \sigma(a_2) \leftarrow a_2 = W_2 h_1 + b_2$$

$$\downarrow \hat{y} = \text{softmax}(a_3) \rightarrow L(W, b) = - \sum_{c=1}^3 y_c \log \hat{y}_c$$

Influence of w_{111} : $w_{111} \rightarrow a_1 \rightarrow h_1 \rightarrow a_2$

$$L \leftarrow \hat{y} \leftarrow a_3 \leftarrow h_2$$

$\downarrow$

$$\therefore \frac{\partial L}{\partial w_{111}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial a_3} \times \frac{\partial a_3}{\partial h_2} \times \frac{\partial h_2}{\partial a_2} \times \frac{\partial a_2}{\partial h_1} \times \frac{\partial h_1}{\partial a_1} \times \frac{\partial a_1}{\partial w_{111}}$$